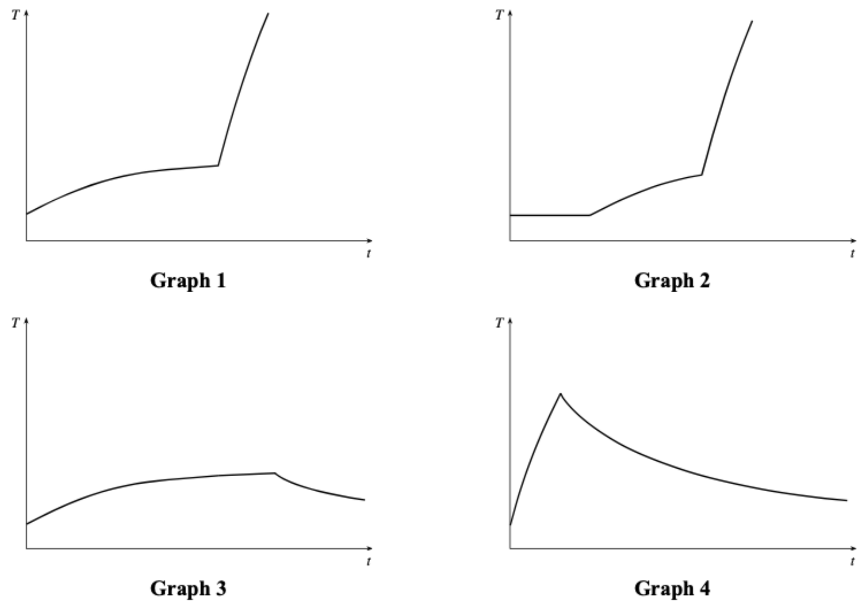
L1.1a Functions & Catalog

Work in your group. Check your answers against the Quarto tonight.

1 · What a function is

1. Four graphs of temperature versus time, and three stories. Match each story to its graph, then **write a story** for the graph that is left over.



(a) I took my roast beef out of the freezer at noon, and left it on the counter to thaw. Then I cooked it in the oven when I got home.

(b) I took my roast beef out of the freezer this morning, and left it on the counter to thaw. Then I cooked it when I got home.

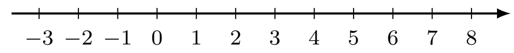
(c) I took my roast beef out of the freezer this morning, and left it on the counter to thaw. I forgot about it, and went out for Chinese food on my way home from work. I put it in the refrigerator when I finally got home.

2. Which of these define y as a function of x? Name the test you used.

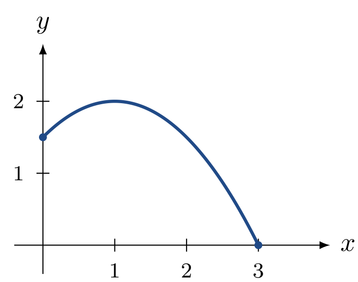
$$x^2 + y^2 = 9 \qquad\qquad y = |x - 1| \qquad\qquad x = y^2 + 1$$

2 · Domain & range

3. $A = [-2, 5)$ and $B = (1, 8]$. Sketch both on the number line, then find $A \cap B$ and $A \cup B$.



4. From the graph, find $f(1)$, $f(3)$, D, and R. **Both notations.**



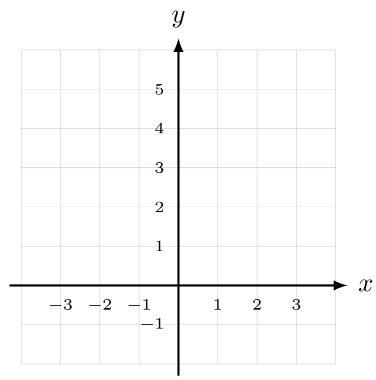
5. $f(x) = \sqrt{x+2}$ — find D and R. **Both notations.**

6. $g(x) = \dfrac{1}{x^2 - x}$ — find `Z`, `VA`, `HA`, sign-probe at -1, 0.5, and 10, then state D.

> **Stop.** Before you probe — where can a rational function fail to be defined, and what does that do to D?

3 · Piecewise & absolute value

7. Sketch $f(x) = \begin{cases} x^2 & x < 0 \\ 1 + x & x > 0 \end{cases}$ and state D and R.



8. Write $|2x + 1|$ as a piecewise function.

Stop. How many places does $2x + 1$ go? Count them before you write anything down.

4 · Symmetry

9. Even, odd, or neither? Show the algebra every time.

(a) $x^5 + x$ (b) $1 - x^4$ (c) $x - x^2$ (d) $x \cos x$

5 · The difference quotient

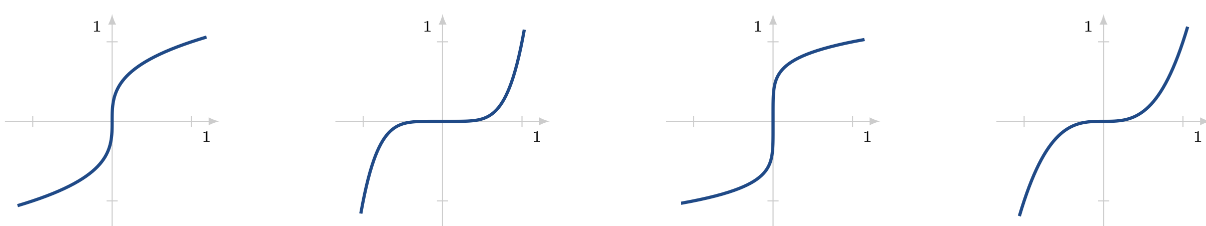
10. $f(x) = x^2 + x$. Find the difference quotient.

Stop. Write the difference quotient down from memory before you substitute anything.

11. $f(x) = 4x - 1$. Find the difference quotient. What is different about this answer, and why?

6 · The function families

12. The four graphs below are x^3, x^5, $\sqrt[3]{x}$, and $\sqrt[5]{x}$ — but which is which? How did you tell the two power functions apart?



A **B** **C** **D**

13. Factor $3x^2 + 2x - 1$ with **Loh**.

14. For $\dfrac{x^2 - x - 6}{x^2 - 4}$ find `Z`, `VA`, `HE`, `HA`, and D.

15. For $y = 2^x + 1$ give `HA`, D, and R.

At the boards

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16. BOARDS $f(x) = \dfrac{1}{2x} + \dfrac{1}{x+1}$

(a) Simplify f into a single rational function and state D.

(b) Find the difference quotient of f.

(c) Put $h = 0$ into the quotient *before* you simplified it, then into your simplified answer. Why do the two disagree?